الاسم والشهرة:----------

الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني
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الامتحانات الرسمية لشهادة الامتياز الفني
لدورة عام 2015 الأولى

رقم المرشح :----------

الاختصاص والرمز : الانظمة والشبكات (23204)
المادة : أنظمة استثمار الجهاز
المستندات المسموح بها : لا شيء

المعدل: (14)
المدة : ساعتان

Treat all the followings questions:

Exercise 1: MCQS multiple choice questions (3.5 pts = 0,5pt/question)
One answer per question is authorized. There's no negative point for a wrong answer.

1. User programs communicate with the operating system and request services (e.g. reading of a record in a file with the file read operation) by:
   A. Interrupts.
   B. System calls.
   C. In the case of device management, the need for a driver program to directly communicate with the device, this service is not supported by the operating system.
2. The interrupt handling mechanism enables:
   A. The user program to take control of the bus.
   B. To allocate an execution priority to the user program.
   C. To interrupt the running of the processor to request a specific job.
3. A compiler is software that produces a code directly executable by operating system
   A. True.
   B. False.
4. With a dual processor server, it is possible to actually run in parallel at a time t in the context of pseudo-parallelism:
   A. One process.
   B. Two processes.
   C. Four processes.
   D. Any number of processes.
5. States of the processor: what transition is triggered by the event of the arrival of a character input by the keyboard?
   A. Active-> blocked.
   B. Blocked-> ready.
   C. Ready-> active.
6. A page fault occurs:
   A. When the virtual page is not present in physical memory.
   B. When the virtual page can no longer be accessed on the disk due to an inaccessible sector.
   C. When the page loaded in real memory is well-presented but unexploitable.
7. The virtual memory paging system is a mechanism of management of main memory that allows:
   A. To significantly increase the number of processes in main memory.
   B. To increase the space of physical memory of the machine.
   C. To increase only the shared memory space between processes.

Exercise 2: (7.5 pts / 1.5 pts par question)

Question 1: Give the definition of a deadlock and specify the difference between a deadlock and a lock.

Question 2: What is a critical section and explain its relationship with the mutual exclusion.

Question 3: Explain the principle of paging.

Question 4: What are the different states of a process? Specify the transitions shift from one state to another.

Question 5: What is the role of the FAT in the management of files? How does MS-DOS manage files?

Exercise 3: A store can accommodate a limited number of customers. This limit is represented by the number N of trolley available at the entrance of the store. A customer who arrives: wait if there is no available trolley. When a customer acquires a trolley, he enters the store to make purchases. As soon as he finishes, he releases his trolley when going out of the store.
You can simulate clients to parallel processes and trolleys to shared resources.

1) Write the algorithm of each customer. **(1 pt)**
2) Consider now that there are two categories of customers: "Subscribers" and "non-subscribers". There is no mutual exclusion between subscribers and non-subscribers; however subscribers have priorities for the acquisition of trolleys.
Write the algorithms of the processes "Subscribers" and "non-subscribers". **(2 pts)**

Exercise 4: Consider the following set of processes, with the duration of the cycle of CPU in ms:

Process	Duration of the cycle	Priority
P1	5	3
P2	3	1
P3	1	4
P4	7	3
P5	2	2

It is assumed that the processes arrive in the following order P1, P2, P3, P4, and P5 at time 0.

1) Draw four Gantt charts illustrating the implementation of these processes by using the algorithms for scheduling FCFS (first come first served), SJF (shortest job first), a priority without requisition (smallest number indicates highest priority) and RR (round-robin with a quantum of 2). **(2 pts)**
2) What is the average time for restitution of a process for each of the scheduling algorithms? **(1 pt)**

Exercise 5: Given the following table of pages:

Logical page number	Physical page number
0	4
1	6
2	8
3	9
4	12
5	1

Knowing that the virtual and physical pages are 1 k Bytes, what is the memory address corresponding to each of the following virtual addresses: 5162 and 2801? **(3 pts)**